



Semiconducting Borophene Realized via Hydrogenation-Driven Structural Reconstruction

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Abstract: Hydrogenation provides a robust approach to both dramatically improve borophene's ambient stability and induce semiconducting behavior—two essential requirements for its integration into nanoelectronic and optoelectronic devices. However, the vast configurational diversity of hydrogen adsorption patterns impedes the identification of a definitive semiconducting hydrogenated borophene phase. In this work, we conducted systematic high-throughput searches to investigate various hydrogen coverages on α' -borophene and determined that semi-hydrogenated configurations constitute the most promising semiconducting candidates. Guided by these predictions, we synthesized semi-hydrogenated borophene, α' -B₈H₄ via chemical vapor deposition. Subsequent extensive structural characterization and complementary computational simulations provided compelling evidence for the formation of this stable semiconducting phase. In sharp contrast to the previously proposed α' -4H model, the synthesized α' -B₈H₄ undergoes hydrogenation-induced structural reconstruction—stabilized by both two-center–two-electron B–H and three-center–two-electron B–H–B bonds—whose ensuing out-of-plane buckling drives strong hybridization between in-plane and out-of-plane *p* orbitals, thereby opening a band gap. This combined theoretical and experimental study identifies an air-stable semiconducting borophene and paves the way for borophenes as next-generation electronic and optoelectronic materials.

Introduction

Borophene has attracted considerable interest due to its exceptional properties, including anisotropic metallic-ity, high mechanical strength and flexibility, broad polymorphism, phonon-mediated superconductivity, optoelec-

tronic anisotropy, optical transparency, and ultrahigh thermal conductivity.^[1–15] Contrary to graphene, whose two-dimensional (2D) form is readily obtained by exfoliation from graphite, borophene has no bulk layered analogue.^[16] Instead, the electron-deficient nature of boron enables its controlled epitaxial deposition on a variety of metal substrates (e.g., Ag(111), Au(111), Cu(111), Al(111), Ir(111), Ru(0001)) under ultrahigh-vacuum conditions.^[17–34] The resultant borophene polymorphs—such as α , α' , β_{12} , β_{13} , and χ_3 , comprise triangular boron lattices with periodic hexagonal vacancies to satisfy the octet rule^[35] and possess an inherently metallic electronic structure, thereby impeding their integration into nanoelectronic and optoelectronic devices. Another critical challenge is borophene's poor ambient stability^[17] as it undergoes rapid degradation outside ultrahigh-vacuum conditions, severely hindering its incorporation into practical devices.

Hydrogenation has been demonstrated to simultaneously improve the chemical stability and tune the properties of various 2D materials.^[36–45] Experimental studies have demonstrated that hydrogenation reduces the ambient oxidation rate of the $\nu_{1/5}$ and $\nu_{1/6}$ borophene phases by over two orders of magnitude, thereby substantially boosting their environmental stability.^[46] Nevertheless, despite hydrogenation, these polymorphs retain the metallic electronic character of their parent borophene phases.^[46] A semi-hydrogenated α' -borophene—identified as the theoretically predicted α' -4H structure—was synthesized experimentally.^[47] This phase not only exhibits semiconducting behavior but also demonstrates exceptional chemical resilience, retaining stability in strongly acidic and basic environments,

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which has enabled its incorporation into prototype device architectures.^[47–49]

The crystal structure constitutes the fundamental descriptor of any 2D material, directly governing its physical and chemical behavior; consequently, its unambiguous determination is essential for elucidating the relationship between experimental observations and intrinsic material properties. Yet, hydrogen's low scattering power generally prevents direct localization by X-ray diffraction and electron microscopy, so theoretical guidance is essential for reliable structural assignment.^[50–58] Nevertheless, the enormous configurational degrees of freedom in hydrogen adsorption, arising from boron's multicenter B–B bonding, pose a significant challenge to the accurate determination of hydrogenated borophene structures.

In this work, by integrating high-throughput first-principles screening with comprehensive experimental characterization, we discover a stable semi-hydrogenated α' -borophene phase, referred to as α' -B₈H₄, that intrinsically exhibits semiconducting behavior. Markedly distinct from the previously proposed α' -4H model, α' -B₈H₄ undergoes hydrogenation-induced structural reconstruction stabilized by two-center–two-electron (2c–2e) B–H bonds and three-center–two-electron (3c–2e) B–H–B interactions. This reconstruction induces out-of-plane buckling that fosters strong hybridization between in-plane and out-of-plane *p* orbitals, thereby opening a band gap in the otherwise metallic α' -borophene.

Results and Discussion

First-Principles Screening

We began by randomly generating hydrogenated configurations through the placement of hydrogen atoms on the fixed unit cell of α' -borophene, with various B–H ratios (B₈H_{*x*}, *x* = 1–6), using the CALYPSO method.^[59–64] The hydrogen atoms were positioned at top, bridge, and hollow sites. Up to 2000, distinct configurations were generated for each ratio. All candidate structures were fully optimized, and a subset of low-energy configurations was selected for electronic structure calculations. These calculations reveal that the semi-hydrogenated B₈H₄ composition is the optimal hydrogenation ratio for achieving semiconducting behavior.

To further comprehensively explore the configurational diversity of B₈H₄, we systematically enumerated all possible hydrogen adsorption patterns, yielding approximately 8500 distinct structures (Figure S1a, left panel). Only top and bridge sites were considered as hydrogen adsorption at hollow sites was found to be unstable. Among the enumerated configurations, dozens of structures were identified that are energetically more favorable than the previously proposed α' -4H model (Figure S1a, right panel). After removing redundant and symmetry-equivalent structures, 12 distinct phases were obtained (Figure 1a). Notably, the ground state structure is energetically more favorable than the α' -4H phase by approximately 69 meV/atom.

We examined the effect of hydrogen coverage and adsorption site preferences on the relative energetics of α' -borophene (Figures S1b,c). Among these low-energy structures, symmetric adsorption patterns, such as two hydrogen atoms on each side (2H top / 2H bottom), occur more frequently than asymmetric ones, such as 1H top / 3H bottom or 3H top / 1H bottom (Figure 1b). In contrast, configurations in which all four hydrogen atoms are adsorbed on the same surface consistently exhibit higher relative energies. Unexpectedly, the ground-state structure adopts a 3H top / 1H bottom configuration. We also observe that configurations with all four hydrogen atoms, or three out of four, adsorbed at bridge sites are considerably less stable than those involving only top-site or mixed-site adsorption (Figure 1c).

Crystal Structure of α' -B₈H₄

Figure 2a,b depict the crystal structure of the stable semi-hydrogenated borophene phase, referred to as α' -B₈H₄, which crystallizes in the *Cm* space group. The optimized lattice parameters are *a* = *b* = 4.54 Å and γ = 101.4°. The B–B bond lengths range from 1.64 to 1.79 Å. Three hydrogen atoms are adsorbed on top of the B3, B4, and B5 atoms, forming the 2c–2e B–H bonds, while an additional hydrogen atom bridges the B1 and B2 atoms to form a 3c–2e B1–H–B2 bond (Figure 2a). The average bond lengths for the B–H top-site and B–H–B bridge-site configurations are 1.20 and 1.34 Å, respectively.

The 3c–2e B–H–B bond exhibits weak hybridization between B 2*p_z* and H 1*s* orbitals, as evidenced by minimal orbital overlap (Figure S2; top panel) and delocalized bonding interactions (Figure S3a). This bonding character helps maintain local planarity (Figure 2b, top panel). In contrast, the 2c–2e B–H bonds involve strong B 2*p_z*–H 1*s* hybridization, indicated by significant orbital overlap (Figure S2; bottom panel) and sharply localized electron density (Figure S3b) between B and H atoms, leading to a pronounced out-of-plane buckling of Δz = 2.04 Å. This buckling behavior is consistent with trends observed in other hydrogenated borophene phases.^[46,50,58]

These bonding motifs and geometric distortions indicate that α' -B₈H₄ undergoes a hydrogenation-induced structural reconstruction, distinct from α' -4H, where hydrogenation preserves the borophene lattice. Specifically, the reconstruction involves breaking the original B2–B3 bond of pristine α' -borophene and forming a new B1–B2 bond (1.74 Å), as illustrated in Figure 2c. Electron localization function analysis supports this conclusion, showing the disappearance of localized electron density between B2 and B3 and the emergence of new localized electron density between B1 and B2 (Figure 2d).

Consequently, B5 becomes six-coordinated, forming a pentagonal pyramidal geometry with an average $\angle$ H–B–B bond angle of 126.18° (Figure 2b, bottom panel). Meanwhile, B3 and B4 become five-coordinated, adopting open pyramidal configurations with average $\angle$ H–B–B angles of 122.70°. These angles are significantly larger than the ideal 109.5° expected for *sp*³ hybridization, highlighting the complex

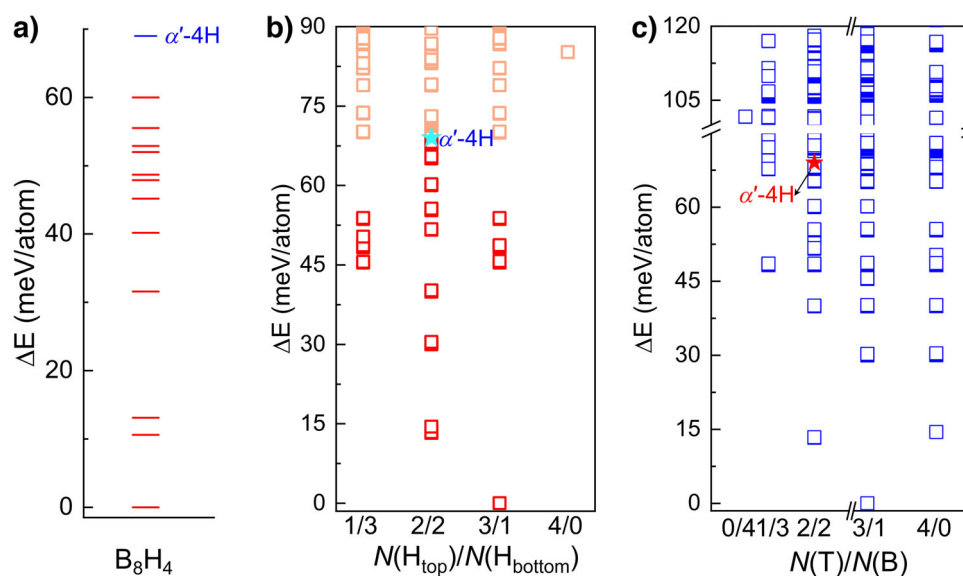


Figure 1. Relative energy analysis of semi-hydrogenated α' -borophene structures. a) Relative energies of various configurations referenced to the ground-state structure with redundant or structurally similar variants removed; the α' -4H phase is indicated by blue short lines. b) Relative energy distribution as a function of the number of H atoms adsorbed on the top (H_{top}) and bottom (H_{bottom}) surfaces. c) Relative energy distribution as a function of the number of H atoms adsorbed on top (T) and bridge (B) sites.

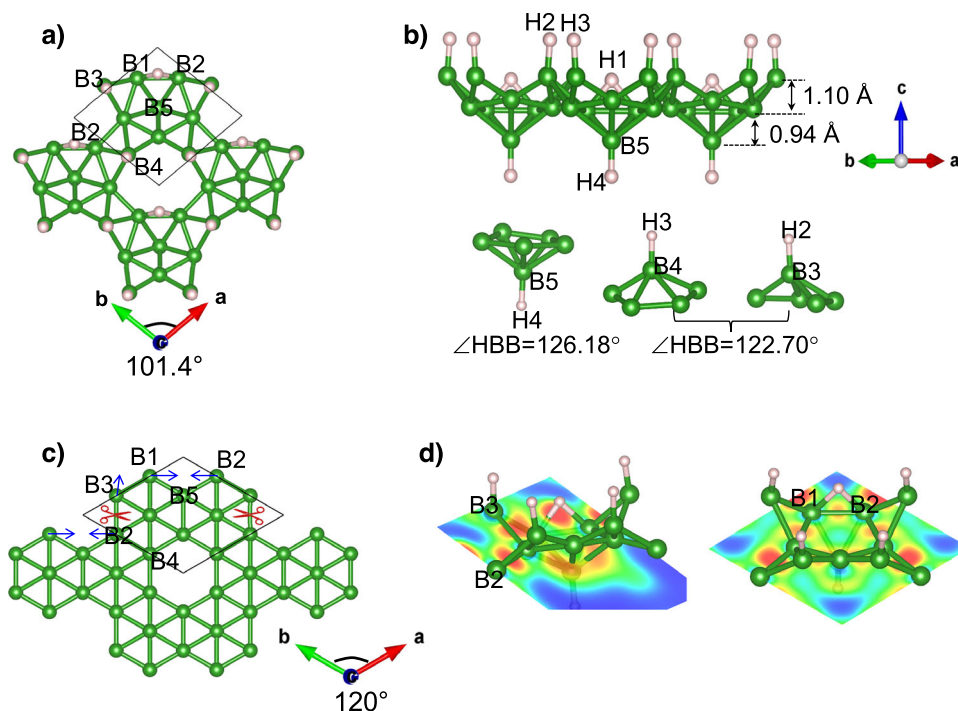


Figure 2. Evolution of crystal structure of global minimum semi-hydrogenated α' -borophene structure. a) Top view of the optimized α' - B_8H_4 . b) Side view of α' - B_8H_4 , highlighting the out-of-plane buckling and local structural units. c) Top view of pristine α' -borophene. d) 2D electron localization function maps of α' - B_8H_4 , illustrating bonding rearrangements after hydrogenation.

and unconventional bonding environment in hydrogenated borophene.

The hydrogenation-induced reconstruction yields a higher adsorption energy for α' - B_8H_4 than for phases that largely preserve the pristine borophene lattice (Table 1), indicating that the bonding energy gained from reconstruction out-

weighs the lattice-distortion cost and thus favors a more stable hydrogenated phase. The cohesive energy of α' - B_8H_4 is -4.30 eV/atom relative to isolated B atoms and H_2 molecules, confirming its thermodynamic favorability. Phonon calculations show no imaginary modes across the Brillouin zone (Figure S4), evidencing dynamical stability.

Table 1: Calculated adsorption energies (E_{ad} in unit of eV/atom) of α' - B_8H_4 , α' -4H and other hydrogenated borophene structures.

2D structures	E_{ad} (eV/atom)	References
α' - B_8H_4	-1.25	This work
α' -4H	-1.18	[47]
$\nu_{1/6}$ -H _{top}	-0.40	[46]
$\nu_{1/6}$ -H _{bridge}	-0.58	[46]
$\nu_{1/6}$ -H _{top} + H _{bridge} (C _{0.4} - $\nu_{1/6}$)	-0.90	[46,50]
C _{0.3} - $\nu_{1/6}$	-0.78	[50]

Ab initio molecular dynamics simulations at 300 K reveal no significant structural degradation, demonstrating thermal stability (Figure S5). The calculated elastic constants satisfy the Born stability criteria for 2D systems,^[65] indicating mechanical stability (Table S1).

Synthesis, Structural Characterization, and Spectroscopy Measurements of α' - B_8H_4

We successfully synthesized large-area borophene nanosheets on aluminum foil via in situ thermal decomposition of NaBH₄, using hydrogen as the carrier gas under a programmed temperature profile (Figure S6). Details regarding the reproducibility of our synthesis and the controlled experimental conditions are provided in the Supporting Information (Figures S7 and S8). X-ray photoelectron spectroscopy (XPS) measurements show that the hydrogenated borophene layer is physically adsorbed on the aluminum substrate without forming detectable Al-B chemical bonds (Figure S9).

High-resolution transmission electron microscopy (HRTEM) image confirmed that the hydrogenated borophene is single crystalline (Figure 3a). A magnified HRTEM image extracted from the orange rectangular region in Figure 3a was reconstructed by masking the 2D fast Fourier transform pattern (Figure 3b), revealing a rhomboidal lattice with one bright protrusion per unit cell. The measured lattice spacing (4.40 Å) and intersection angle (75.2°) are in good agreement with the theoretically predicted structure of α' - B_8H_4 . Furthermore, the simulated TEM image based on the α' - B_8H_4 model (Figure 3c) closely reproduces the experimental HRTEM features, providing the support to the proposed structural model.

The selected-area electron diffraction (SAED) pattern was acquired from the same region as TEM, showing distinct diffraction spots with rhombohedral symmetry, indicative of high crystallinity (Figure 3d; top panel). The simulated SAED pattern (Figure 3d; bottom panel), derived from the α' - B_8H_4 model, reproduces the experimental spot positions and symmetry. These structural characterizations collectively provide compelling evidence for the successful synthesis of the predicted α' - B_8H_4 phase.

The comparison of B 1s XPS spectra of α' - B_8H_4 before and after one week of exposure to air confirms its good ambient stability, as evidenced by nearly unchanged characteristic peaks with only slight broadening and minor shifts in binding energy (Figure S10). To further assess chemical robustness, the sample was immersed in 1 mol L⁻¹ sodium

hydroxide, 1 mol L⁻¹ hydrochloric acid, and 5% hydrofluoric acid solutions for 48 h, followed by centrifugation to recover the sheets for XPS characterization (Figure S11). The primary spectral features remain intact in all cases, with only negligible signs of degradation, demonstrating the remarkable chemical stability of α' - B_8H_4 .

Raman spectroscopy measurements show that the vibrational frequencies of B-H bonds reach up to 2528 cm⁻¹, significantly higher than those of B-B bonds, which lie below 1200 cm⁻¹ (Figure 3e). Similar hydrogenation-induced bond strengthening has also been observed in other borophene phases.^[47,55] The point group of α' - B_8H_4 at the Γ point is Cs. The vibrational representation at Γ point is $\Gamma_{vib} = 24A' + 12A''$, including acoustic modes $\Gamma_{acoustic} = 2A' + A''$ and the optical modes $\Gamma_{optical} = 18A' + 15A''$ (Table S2). Owing to the absence of inversion symmetry, both A' and A'' are Raman active. As shown in Figure 3e, the calculated Raman modes show agreement with the experimentally measured data. In particular, the peak near 2528 cm⁻¹ arises from the vibration of 2c-2e B-H bonds, corresponding to out-of-plane vibrations involving the B5-H, B3-H, and B4-H bonds (Figures 2b,c and S12). The features observed below 1200 cm⁻¹ are attributed to various B-B bond vibrations.

Although theoretical simulations predict two vibrational modes near 1990 and 1550 cm⁻¹ corresponding to the 3c-2e B-H-B bond (Figure S12), these peaks are not clearly observed in the experimental Raman spectra. This discrepancy may arise from the intrinsically low Raman activity of these modes or the low abundance of B-H-B bonds within the structure. Overall, the observed Raman features are consistent with the theoretical vibrational fingerprints of B-B and B-H bonding, providing strong evidence that the synthesized hydrogenated borophene corresponds to the α' - B_8H_4 phase.

The optical properties of hydrogenated borophene were investigated using photoluminescence (PL) spectroscopy (Figure 3f). A pronounced emission peak was observed at 493 nm, corresponding to a photon energy of 2.52 eV. The PBE0 hybrid functional has been shown to provide reliable bandgap predictions for borophene systems, often yielding results in closer agreement with experimental measurements.^[51] Therefore, we employed the PBE0 functional to investigate the electronic properties of the α' - B_8H_4 phase. The results reveal that α' - B_8H_4 is a semiconductor with an indirect band gap of 2.54 eV (Figures 3g and S13), in excellent agreement with the experimental value. The projected density of states shows that the valence band maximum is primarily composed of B p_x orbitals, while the conduction band minimum is dominated by B p_z orbitals. Moreover, a significant hybridization between the p_z and $p_{x,y}$ orbitals is evident in both the occupied and unoccupied states.

Mechanism of the Semiconducting Behavior of α' - B_8H_4

To elucidate the underlying mechanism of the semiconducting behavior in α' - B_8H_4 , we calculated the orbital-resolved band structures of pristine α' -borophene and hydrogenated α' - B_8H_4 (Figure 4). In α' -borophene, a triply degenerate state

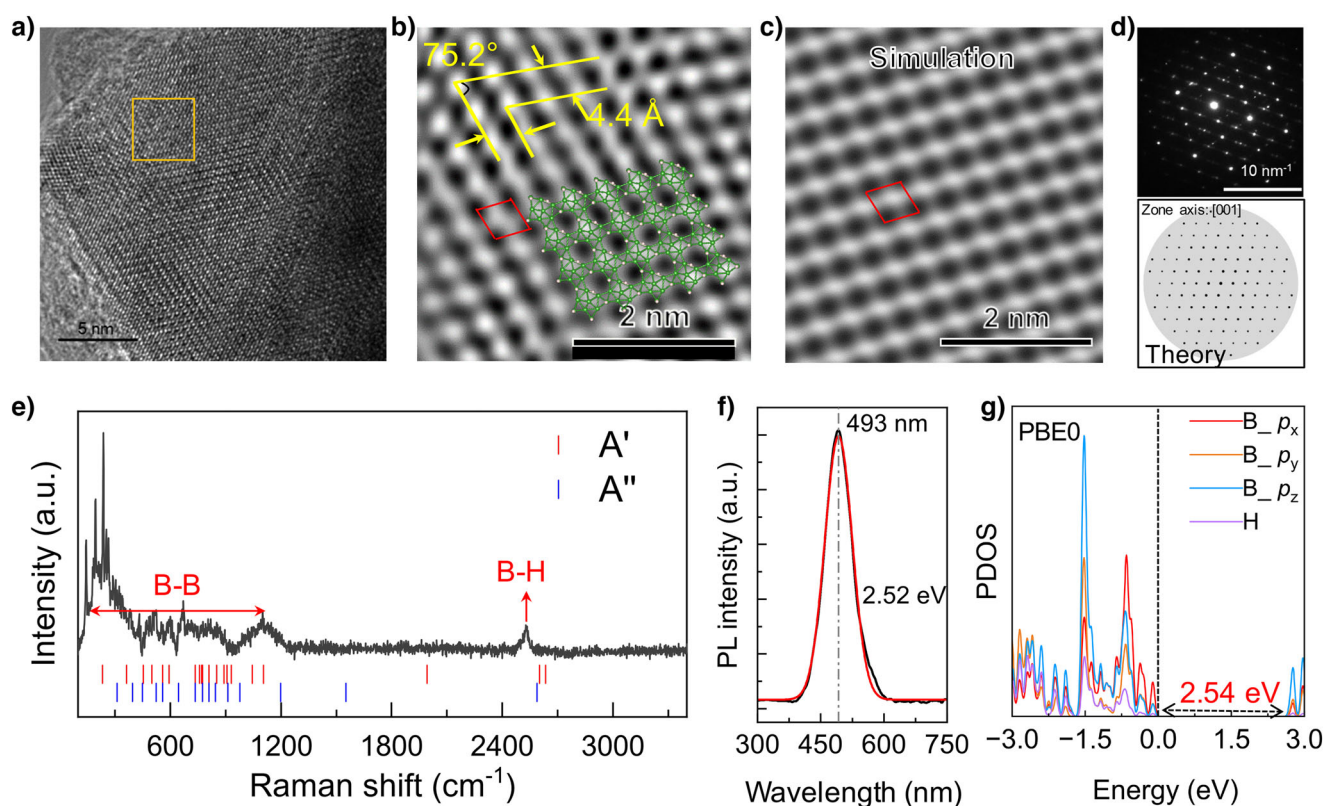


Figure 3. Characterization and simulation of the hydrogenated α' -borophene phase. a) Experimental HRTEM image of the hydrogenated α' -borophene structure. b) HRTEM image by masking the fast Fourier transform pattern extracted from the orange rectangular region in (a). c) Simulated HRTEM image based on the α' -B₈H₄ structure. d) Experimental and simulated SAED patterns. e) Comparison of experimental and calculated Raman spectra; vertical lines mark the computed peak positions of the Raman active A' and A'' modes at the Γ point. f) PL spectrum displaying a characteristic emission peak. g) Projected density of states (PDOS) calculated using the PBE0 hybrid functional.

appears at the Γ point near the Fermi level, primarily derived from B p_z orbitals (Figure 4a). Due to the nearly planar geometry of α' -borophene, the p_z orbitals are orthogonal to the in-plane p_x and p_y orbitals, resulting in minimal orbital hybridization.

Upon hydrogenation, significant structural buckling is induced, with several boron atoms displaced out of the plane due to the B–H interactions (Figure 2b). This distortion breaks the orbital orthogonality, enabling hybridization between p_z and $p_{x/y}$ orbitals (Figure 4b). The resulting orbital interaction lifts the degeneracy and induces a bandgap in α' -B₈H₄. To further confirm the essential role of buckling-induced orbital hybridization, we computed the band structure of a hypothetical flat α' -B₈H₄ structure (Figure S14). The absence of p_z - $p_{x/y}$ hybridization and the persistence of metallic nature in this model confirm that the semiconducting behavior of α' -B₈H₄ originates from hydrogenation induced orbital interactions mediated by lattice buckling.

Beyond α' -B₈H₄, we identified ten additional phases that are both dynamically stable and energetically more favorable than α' -4H (Figures S15 and S16). These structures retain lattice geometries closely resembling that of α' -4H, with varying degrees of out-of-plane buckling. As a result, their simulated TEM images are nearly indistinguishable from one another (Figure S17). Notably, nine of the ten low-energy configurations exhibit semiconducting behavior, with

several displaying band gaps comparable to that of the α' -4H phase (Figures S18 and S19). The semiconducting behavior of these newly identified semi-hydrogenated borophene phases originates from the same mechanism as that of the α' -B₈H₄ phase. These findings raise important questions regarding the structural assignment of the -4H phase and suggest that a re-evaluation or refinement of its experimental interpretation may be warranted.

Conclusion

Integration of high-throughput first-principles screening with chemical vapor deposition has led to the discovery of an air- and chemically stable semi-hydrogenated α' -borophene phase, α' -B₈H₄, which exhibits intrinsic semiconducting behavior. Experimental verification via the high-resolution transmission electron microscopy, the selected-area electron diffraction, Raman spectroscopy, and photoluminescence spectroscopy confirm the structure of the synthesized phase, featuring a hydrogenation-driven structural reconstruction stabilized by two-center-two-electron B–H and three-center-two-electron B–H–B bonding motifs. Electronic structure analysis shows that hydrogenation-induced out-of-plane buckling fosters strong hybridization between B $2p_z$ and

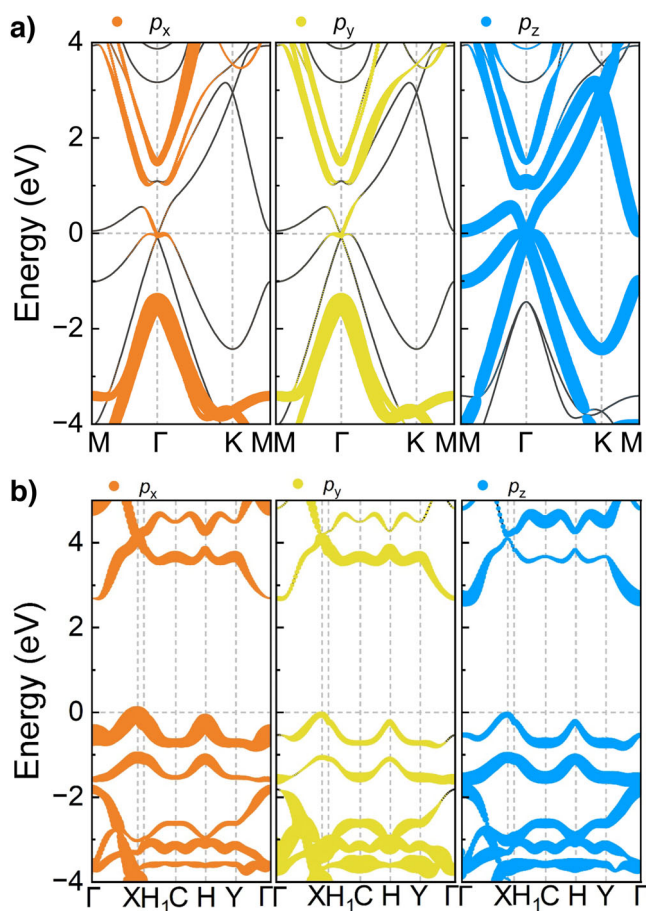


Figure 4. Origin of the semiconducting nature of α' -B₈H₄. Orbital-decomposed band structures of a) pristine α' -borophene and b) hydrogenated α' -B₈H₄. The projected contributions from B p_x , p_y , and p_z orbitals are shown.

$2p_{x/y}$ orbitals, opening a bandgap of ~ 2.52 eV. These results not only deepen our understanding of chemically engineered semiconducting borophene phases but also establish a versatile approach for designing two-dimensional materials tailored for nanoelectronic and optoelectronic applications.

Supporting Information

The authors have cited additional references within the Supporting Information.^[17,66–85]

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords: Chemical vapor deposition • First-principles calculations • Semiconducting borophene • Semi-hydrogenation • Structural reconstruction

- [1] A. J. Mannix, Z. Zhang, N. P. Guisinger, B. I. Yakobson, M. C. Hersam, *Nat. Nanotechnol.* **2018**, *13*, 444–450, <https://doi.org/10.1038/s41565-018-0157-4>.
- [2] C. Hou, G. Tai, Y. Liu, Z. Wu, X. Liang, X. Liu, *Nano Research Energy* **2023**, *2*, e9120051.
- [3] L. Adamska, S. Sadasivam, J. J. Foley IV, P. Darancet, S. Sharifzadeh, *J. Phys. Chem. C* **2018**, *122*, 4037–4045.
- [4] Z. Wu, C. Shifan, Z. Wu, Y. Liu, W. Shao, X. Liang, C. Hou, G. Tai, *Nano Res.* **2024**, *17*, 3053–3060.
- [5] E. S. Penev, A. Kutana, B. I. Yakobson, *Nano Lett.* **2016**, *16*, 2522–2526, <https://doi.org/10.1021/acs.nanolett.6b00070>.
- [6] M. Gao, Q.-Z. Li, X.-W. Yan, J. Wang, *Phys. Rev. B* **2017**, *95*, 024505.
- [7] Z. Zhang, Y. Yang, E. S. Penev, B. I. Yakobson, *Adv. Funct. Mater.* **2017**, *27*, 1605059.
- [8] H. Tang, S. Ismail-Beigi, *Phys. Rev. Lett.* **2007**, *99*, 115501.
- [9] X. Wu, J. Dai, Y. Zhao, Z. Zhuo, J. Yang, C. Zeng, *ACS Nano* **2012**, *6*, 7443–7453.
- [10] Z. Zhang, Y. Yang, G. Gao, B. I. Yakobson, *Angew. Chem. Int. Ed.* **2015**, *54*, 13022–13026; *Angew. Chem.* **2015**, *127*, 13214–13218.
- [11] E. S. Penev, S. Bhowmick, A. Sadrzadeh, B. I. Yakobson, *Nano Lett.* **2012**, *12*, 2441–2445.
- [12] S.-G. Xu, X.-T. Li, Y.-J. Zhao, J.-H. Liao, W.-P. Xu, X.-B. Yang, H. Xu, *J. Am. Chem. Soc.* **2017**, *139*, 17233–17236.
- [13] Y.-L. Zhang, J.-H. Yang, H. Xiang, X.-G. Gong, *J. Phys. Chem. Lett.* **2020**, *12*, 576–584.
- [14] S. Guha, A. Kabiraj, S. Mahapatra, *ACS Appl. Mater. Interfaces* **2023**, *15*, 3182–3191.
- [15] C. Hou, G. Tai, Y. Liu, R. Liu, X. Liang, Z. Wu, Z. Wu, *Nano Energy* **2022**, *97*, 107189.
- [16] X. Sun, X. Liu, J. Yin, J. Yu, Y. Li, Y. Hang, X. Zhou, M. Yu, J. Li, G. Tai, W. Guo, *Adv. Funct. Mater.* **2017**, *27*, 1603300.
- [17] J. Mannix, X.-F. Zhou, B. Kiraly, J. D. Wood, D. Alducin, B. D. Myers, X. Liu, B. L. Fisher, U. Santiago, J. R. Guest, M. J. Yacaman, A. Ponce, A. R. Oganov, M. C. Hersam, N. P. Guisinger, *Science* **2015**, *350*, 1513–1516, <https://doi.org/10.1126/science.aad1080>.
- [18] G. Tai, T. Hu, Y. Zhou, X. Wang, J. Kong, T. Zeng, Y. You, Q. Wang, *Angew. Chem. Int. Ed.* **2015**, *54*, 15473–15477.

- [19] B. Feng, J. Zhang, Q. Zhong, W. Li, S. Li, H. Li, P. Cheng, S. Meng, L. Chen, K. Wu, *Nat. Chem.* **2016**, *8*, 563–568, <https://doi.org/10.1038/nchem.2491>.
- [20] Q. Zhong, J. Zhang, P. Cheng, B. Feng, W. Li, S. Sheng, H. Li, S. Meng, L. Chen, K. Wu, *J. Phys.: Condens. Matter* **2017**, *29*, 095002.
- [21] B. Kiraly, X. Liu, L. Wang, Z. Zhang, A. J. Mannix, B. L. Fisher, B. I. Yakobson, M. C. Hersam, N. P. Guisinge, *ACS Nano* **2019**, *13*, 3816–3822.
- [22] Liu, Z. Zhang, L. Wang, B. I. Yakobson, M. C. Hersam, *Nat. Mater.* **2018**, *17*, 783–788.
- [23] W. Li, L. Kong, C. Chen, J. Gou, S. Sheng, W. Zhang, H. Li, L. Chen, P. Cheng, K. Wu, *Sci. Bull.* **2018**, *63*, 282–286.
- [24] L. Zhu, B. Zhao, T. Zhang, G. Chen, S. A. Yang, *J. Phys. Chem. C* **2019**, *123*, 14858–14864.
- [25] R. Wu, I. K. Drozdov, S. Eltinge, P. Zahl, S. Ismail-Beigi, I. Božović, A. Gozar, *Nat. Nanotechnol.* **2019**, *14*, 44–49.
- [26] X.-J. Weng, Y. Zhu, Y. Xu, J. Bai, Z. Zhang, B. Xu, X.-F. Zhou, Y. Tian, *Adv. Funct. Mater.* **2024**, *34*, 2314576.
- [27] R. Wu, S. Eltinge, I. K. Drozdov, A. Gozar, P. Zahl, J. T. Sadowski, S. Ismail-Beigi, I. Božović, *Nat. Chem.* **2022**, *14*, 377–383.
- [28] N. A. Vinogradov, A. Lyalin, T. Taketsugu, A. S. Vinogradov, A. Preobrajenski, *ACS Nano* **2019**, *13*, 14511–14518.
- [29] Y. V. Kaneti, D. P. Benu, X. Xu, B. Yuliarto, Y. Yamauchi, D. Golberg, *Chem. Rev.* **2021**, *122*, 1000–1051, <https://doi.org/10.1021/acs.chemrev.1c00233>.
- [30] X. Liu, Q. Li, Q. Ruan, M. S. Rahn, B. I. Yakobson, M. C. Hersam, *Nat. Mater.* **2022**, *21*, 35–40.
- [31] P. Sutter, E. Sutter, *Chem. Mater.* **2021**, *33*, 8838–8843, <https://doi.org/10.1021/acs.chemmater.1c03061>.
- [32] C. Chen, H. Lv, P. Zhang, Z. Zhuo, Y. Wang, C. Ma, W. Li, X. Wang, B. Feng, P. Cheng, X. Wu, K. Wu, L. Chen, *Nat. Chem.* **2022**, *14*, 25–31, <https://doi.org/10.1038/s41557-021-00813-z>.
- [33] Y. Xu, X. Xuan, T. Yang, Z. Zhang, S.-D. Li, W. Guo, *Nano Lett.* **2022**, *22*, 3488–3494.
- [34] Z. Zhang, E. S. Penev, B. I. Yakobson, *Chem. Soc. Rev.* **2017**, *46*, 6746–6763, <https://doi.org/10.1039/C7CS00261K>.
- [35] Q. Li, E. B. Aklile, A. Tsui, M. C. Hersam, *Nat. Chem.* **2025**, *17*, 642–652, <https://doi.org/10.1038/s41557-025-01773-4>.
- [36] D. C. Elias, R. R. Nair, T. Mohiuddin, S. Morozov, P. Blake, M. Halsall, A. C. Ferrari, D. Boukhvalov, M. Katsnelson, A. Geim, K. S. Novoselov, *Science* **2009**, *323*, 610–613.
- [37] R. Balog, B. Jørgensen, L. Nilsson, M. Andersen, E. Rienks, M. Bianchi, M. Fanetti, E. Lægsgaard, A. Baraldi, S. Lizzit, Z. Sljivancanin, F. Besenbacher, B. Hammer, T. G. Pedersen, P. Hofmann, L. Hornekaer, *Nat. Mater.* **2010**, *9*, 315–319.
- [38] M. Pumera, C. H. A. Wong, *Chem. Soc. Rev.* **2013**, *42*, 5987–5995.
- [39] E. Bianco, S. Butler, S. Jiang, O. D. Restrepo, W. Windl, J. E. Goldberger, *ACS Nano* **2013**, *7*, 4414–4421.
- [40] J. Qiu, H. Fu, Y. Xu, A. Oreshkin, T. Shao, H. Li, S. Meng, L. Chen, K. Wu, *Phys. Rev. Lett.* **2015**, *114*, 126101.
- [41] Y. Jiao, F. Ma, J. Bell, A. Bilic, A. Du, *Angew. Chem. Int. Ed.* **2016**, *55*, 10292–10295; *Angew. Chem.* **2016**, *128*, 10448–10451.
- [42] Y. An, Y. Hou, H. Wang, J. Li, R. Wu, T. Wang, H. Da, J. Jiao, *Phys. Rev. Appl.* **2019**, *11*, 064031.
- [43] D. Li, J. He, G. Ding, Q. Tang, Y. Ying, J. He, C. Zhong, Y. Liu, C. Feng, Q. Sun, H. Zhou, P. Zhou, G. Zhang, *Adv. Funct. Mater.* **2018**, *28*, 1801685.
- [44] B. N. Soskic, J. Bekaert, C. Sevik, M. V. Milosevic, *Nano Lett.* **2024**, *24*, 12650–12657, <https://doi.org/10.1021/acs.nanolett.4c03845>.
- [45] H. Nishino, T. Fujita, N. T. Cuong, S. Tominaka, M. Miyauchi, S. Iimura, A. Hirata, N. Umezawa, S. Okada, E. Nishibori, A. Fujino, T. Fujimori, S. Ito, J. Nakamura, H. Hosono, T. Kondo, *J. Am. Chem. Soc.* **2017**, *139*, 13761–13769, <https://doi.org/10.1021/jacs.7b06153>.
- [46] Q. Li, V. S. C. Kolluru, M. S. Rahn, E. Schwenker, S. Li, R. G. Hennig, P. Darancet, M. K. Chan, M. C. Hersam, *Science* **2021**, *371*, 1143–1148, <https://doi.org/10.1126/science.abg1874>.
- [47] C. Hou, G. Tai, J. Hao, L. Sheng, B. Liu, Z. Wu, *Angew. Chem. Int. Ed.* **2020**, *59*, 10819–10825, <https://doi.org/10.1002/anie.202001045>.
- [48] X. Liang, C. Hou, Z. Wu, Z. Wu, G. Tai, *Nanotechnology* **2023**, *34*, 205701, <https://doi.org/10.1088/1361-6528/acba1e>.
- [49] Z. Wu, X. Liang, Y. Liu, M. Xu, R. Zhu, G. Tai, *Angew. Chem. Int. Ed.* **2025**, *64*, e202416041, <https://doi.org/10.1002/anie.202416041>.
- [50] Y. Xu, P. Zhang, X. Xuan, M. Xue, Z. Zhang, W. Guo, B. I. Yakobson, *J. Phys. Chem. Lett.* **2022**, *13*, 1107–1113, <https://doi.org/10.1021/acs.jpclett.1c03827>.
- [51] M. A. Mohebbpour, S. M. Mozvashi, S. I. Vishkayi, M. B. Tagani, *Phys. Rev. Mater.* **2022**, *6*, 014012, <https://doi.org/10.1103/PhysRevMaterials.6.014012>.
- [52] K. Pang, X. Xu, R. Ku, Y. Wei, T. Ying, W. Li, J. Yang, X. Li, Y. Jiang, *ACS Appl. Nano Mater.* **2022**, *5*, 1419–1425, <https://doi.org/10.1021/acsanm.1c03956>.
- [53] P. Sang, Q. Wang, W. Wei, Y. Li, J. Chen, *ACS Appl. Nano Mater.* **2021**, *4*, 11931–11937.
- [54] Y. Kang, X. Ma, J. Fu, K. Yang, Z. Wang, H. Li, W. Ma, J. Zhang, *J. Phys. Chem. Lett.* **2022**, *13*, 10222–10229, <https://doi.org/10.1021/acs.jpclett.2c02417>.
- [55] M. Yu, Z. Zhang, W. Guo, *J. Phys. Chem. C* **2021**, *125*, 22917–22928, <https://doi.org/10.1021/acs.jpcc.1c07275>.
- [56] M. Zhang, C. He, J. Zhong, *J. Phys. D: Appl. Phys.* **2024**, *57*, 385109, <https://doi.org/10.1088/1361-6463/ad55fc>.
- [57] L.-C. Xu, A. Du, L. Kou, *Phys. Chem. Chem. Phys.* **2016**, *18*, 27284–27289, <https://doi.org/10.1039/C6CP05405F>.
- [58] Z.-Q. Wang, T.-Y. Lu, H.-Q. Wang, Y. P. Feng, J.-C. Zheng, *ACS Appl. Electron. Mater.* **2019**, *1*, 667–674.
- [59] Y. Wang, J. Lv, L. Zhu, Y. Ma, *Phys. Rev. B* **2010**, *82*, 094116.
- [60] Y. Wang, J. Lv, L. Zhu, Y. Ma, *Comput. Phys. Commun.* **2012**, *183*, 2063–2070.
- [61] Y. Wang, M. Miao, J. Lv, L. Zhu, K. Yin, H. Liu, Y. Ma, *J. Chem. Phys.* **2012**, *137*, 224108, <https://doi.org/10.1063/1.4769731>.
- [62] Y. Wang, Y. Ma, *J. Chem. Phys.* **2014**, *140*, 040901, <https://doi.org/10.1063/1.4861966>.
- [63] B. Gao, P. Gao, S. Lu, J. Lv, Y. Wang, Y. Ma, *Sci. Bull.* **2019**, *64*, 301–309.
- [64] X. Shao, J. Lv, P. Liu, S. Shao, P. Gao, H. Liu, Y. Wang, Y. Ma, *J. Chem. Phys.* **2022**, *156*, 014105.
- [65] F. Mouhat, F.-X. Coudert, *Phys. Rev. B* **2014**, *90*, 224104.
- [66] W. Kohn, L. J. Sham, *Phys. Rev.* **1965**, *140*, A1133–A1138, <https://doi.org/10.1103/PhysRev.140.A1133>.
- [67] G. Kresse, J. Hafner, *Phys. Rev. B* **1993**, *48*, 13115–13118, <https://doi.org/10.1103/PhysRevB.48.13115>.
- [68] G. Kresse, J. Furthmüller, *Phys. Rev. B* **1996**, *54*, 11169–11186, <https://doi.org/10.1103/PhysRevB.54.11169>.
- [69] J. P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* **1996**, *77*, 3865–3868, <https://doi.org/10.1103/PhysRevLett.77.3865>.
- [70] P. E. Blöchl, *Phys. Rev. B* **1994**, *50*, 17953.
- [71] H. J. Monkhorst, J. D. Pack, *Phys. Rev. B* **1976**, *13*, 5188–5192, <https://doi.org/10.1103/PhysRevB.13.5188>.
- [72] J. P. Perdew, M. Ernzerhof, K. Burke, *J. Chem. Phys.* **1996**, *105*, 9982–9985, <https://doi.org/10.1063/1.472933>.
- [73] C. Adamo, V. Barone, *J. Chem. Phys.* **1999**, *110*, 6158–6170, <https://doi.org/10.1063/1.478522>.
- [74] J. Heyd, G. E. Scuseria, M. Ernzerhof, *J. Chem. Phys.* **2003**, *118*, 8207–8215, <https://doi.org/10.1063/1.1564060>.
- [75] A. Togo, F. Oba, I. Tanaka, *Phys. Rev. B* **2008**, *78*, 134106, <https://doi.org/10.1103/PhysRevB.78.134106>.

- [76] S. Baroni, S. De Gironcoli, A. Dal Corso, P. Giannozzi, *Rev. Mod. Phys.* **2001**, 73, 515–562, <https://doi.org/10.1103/RevModPhys.73.515>.
- [77] P. Giannozzi, S. Baroni, N. Bonini, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, G. L. Chiarotti, M. Cococcioni, I. Dabo, A. D. Corso, S. D. Gironcoli, S. Fabris, G. Fratesi, R. Gebauer, U. Gerstmann, C. Gougoussis, A. Kokalj, M. Lazzeri, L. Martin-Samos, N. Marzari, F. Mauri, R. Mazzarello, S. Paolini, A. Pasquarello, L. Paulatto, C. Sbraccia, S. Scandolo, G. Sclauzero, A. P. Seitsonen, et al., *J. Phys.: Condens. Matter* **2009**, 21, 395502, <https://doi.org/10.1088/0953-8984/21/39/395502>.
- [78] G. Kresse, D. Joubert, *Phys. Rev. B* **1999**, 59, 1758–1775, <https://doi.org/10.1103/PhysRevB.59.1758>.
- [79] P. Stadelmann, *Microsc. Microanal.* **2003**, 9, 60–61, <https://doi.org/10.1017/S1431927603012224>.
- [80] D. C. Palmer, *Zeitschrift für Kristallographie – Crystalline Materials* **2015**, 230, 559–572.
- [81] S. Takagi, A. Toriumi, M. Iwase, H. Tango, *IEEE Trans. Electron Devices* **1994**, 41, 2357–2362, <https://doi.org/10.1109/16.337449>.
- [82] S. Bruzzone, G. Fiori, *Appl. Phys. Lett.* **2011**, 99, 2108, <https://doi.org/10.1063/1.3665183>.
- [83] G. Fiori, G. Iannaccone, *Proceedings of the IEEE* **2013**, 101, 1653–1669, <https://doi.org/10.1109/JPROC.2013.2259451>.
- [84] A. Rawat, N. Jena, A. D. S. Dimple, *J. Mater. Chem. A* **2018**, 6, 8693–8704, <https://doi.org/10.1039/C8TA01943F>.
- [85] J. Qiao, X. Kong, Z.-X. Hu, F. Yang, W. Ji, *Nat. Commun.* **2014**, 5, 4475, <https://doi.org/10.1038/ncomms5475>.

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